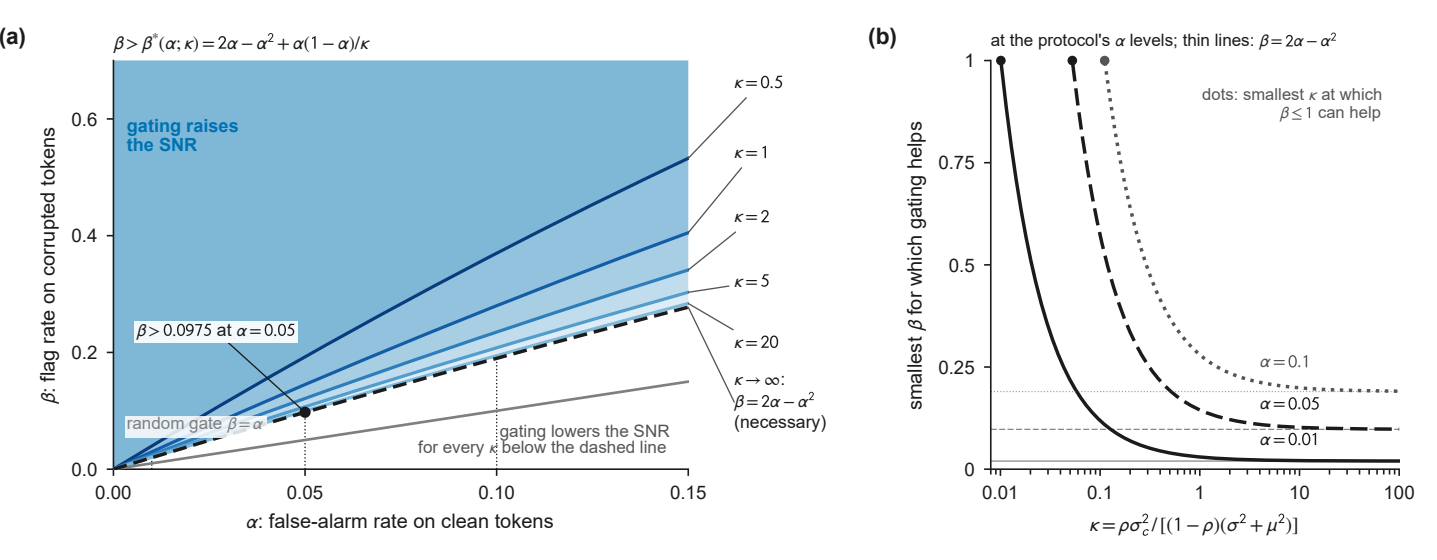

(a)
$\beta > \beta^*(\alpha;\kappa) = 2\alpha - \alpha^2 + \alpha(1-\alpha)/\kappa$
gating raises the SNR
$\beta > 0.0975$ at $\alpha = 0.05$
random gate $\beta = \alpha$
gating lowers the SNR for every κ below the dashed line
$\kappa = 0.5$
$\kappa = 1$
$\kappa = 2$
$\kappa = 5$
$\kappa = 20$
$\kappa \to \infty$: $\beta = 2\alpha - \alpha^2$ (necessary)
β: flag rate on corrupted tokens
α: false-alarm rate on clean tokens
0.0
0.2
0.4
0.6
0.00
0.05
0.10
0.15
(b)
at the protocol's α levels; thin lines: $\beta = 2\alpha - \alpha^2$
dots: smallest κ at which $\beta \le 1$ can help
smallest β for which gating helps
$\alpha = 0.1$
$\alpha = 0.05$
$\alpha = 0.01$
1
0.75
0.5
0.25
0
0.01
0.1
1
10
100
$\kappa = \rho\sigma_c^2 / [(1-\rho)(\sigma^2 + \mu^2)]$